# Structure Preserving Video Augmentation

Ding-Ruei Shen - 1931512
Eindhoven University of Technology, Netherlands

***Abstract*—This research explores various state-of-the-art (SOTA) models for structure-preserving video augmentation. The methods are compared based on their efficiency and effectiveness. Through both qualitative and quantitative evaluations, our findings indicate that RAVE, a diffusion-based model, outperforms ProCST and FDA in style transfer, as evidenced by FVD scores, while ProCST demonstrates superior performance in preserving structural integrity. The main takeaway from this study is that diffusion models hold substantial potential but necessitate fine-tuning to achieve task-specific robustness. Additionally, more diverse evaluation methods, such as alternative metrics or directly testing the synthetic data on downstream tasks, are preferable to ensure reliable assessments and practical applicability.**

***Keywords*—deep learning, domain adaptation, diffusion models, data augmentation, synthetic datasets**

## I. Introduction

The exponential advancements in deep learning have catalyzed transformative changes across multiple fields, including education, robotics, and medicine and healthcare. Among various models, deep generative models, especially Denoising Diffusion Implicit Models (DDIMs) [1], have shown remarkable success in producing high-fidelity images, surpassing conventional GAN-based and VAE-based methods. This ability has extended to generating synthetic datasets for model training, which has numerous benefits, particularly in sensitive fields like medicine. In healthcare, synthetic datasets not only alleviate the shortage of annotated data but also protect patient privacy, as data can be augmented without exposing real patient information. Extensive research corroborates that using synthetic data generated by diffusion models or GAN-based models can significantly improve model performance [2]–[4], as demonstrated in studies [5] where pretraining on synthetic datasets led to improved outcomes when applied to real-world data.

Using synthetically generated X-ray images that retain structural features across frames could substantially enhance the realism of simulated clinical training environments or bolster the diagnostic accuracy of machine learning algorithms. Consequently, novel video augmentation techniques that can adapt to the unique requirements of medical videos by preserving anatomical integrity throughout videos are profoundly necessary. While image augmentation methodologies have achieved significant strides, augmenting medical video data introduces intricate challenges, primarily due to the need to preserve both temporal coherence and spatial fidelity of anatomical structures. Such structure-preserving augmentation is crucial for various applications in diagnostics, therapeutic planning, and simulation-based training for healthcare practitioners [6]–[8]. Despite these advancements, video data augmentation, especially for medical imaging, remains conspicuously underexplored.

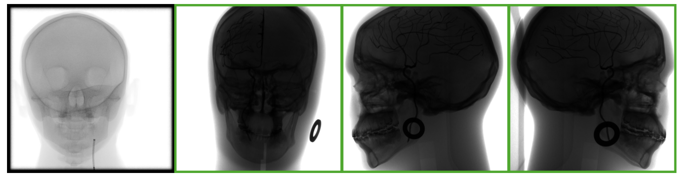

Fig. 1. X-ray images of a head: The leftmost image represents a synthetic (fake) skull, while the other three images are real skulls.

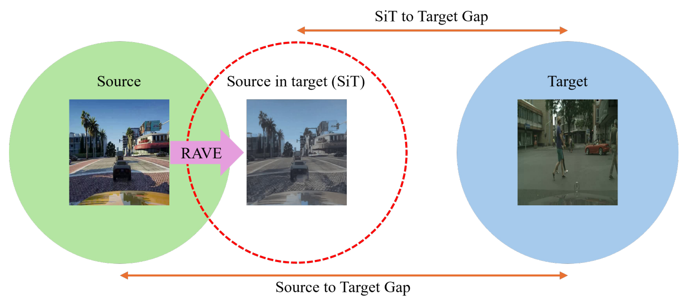


Fig. 2. Depiction of Cross Domain adaption, translating source domain (GTA5) to target domain (Cityscapes) using RAVE.

In this paper, we compare various methods and, based on the results, propose RAVE [9] as a representative model. The primary reason for selecting RAVE is its superior performance across key metrics, such as frame quality, temporal consistency, and textual alignment, as evidenced by quantitative results compared to previous approaches like Text2Video-Zero [10], Rerender [11], TokenFlow [12], and Pix2Video [13]. Although TokenFlow is reported to achieve better structural consistency (measured by WarpSSIM) in their study, RAVE proves to be more effective for longer video editing tasks, successfully handling videos of up to 90 frames. Furthermore, in our quantitative comparisons with non-diffusion-based methods, RAVE consistently demonstrates its superiority, particularly in the context of high-frame-count videos.

For empirical validation, we prepare two different datasets, namely GTA5 (source) and Cityscapes (target), to simulate the domain shift between synthetic and real X-ray data (Fig. 1). GTA5 serves as a proxy for synthetic data and Cityscapes represents real data. This concept is portrayed in (Fig. 2). RAVE provides various text prompts for style guidance, aiming to shrink the domain disparity between GTA5 and Cityscapes. To quantitatively assess performance, we compute the Fréchet Video Distance (FVD) [14] between RAVE-generated outputs and Cityscapes, as well as between the original GTA5 and Cityscapes. Qualitatively, we benchmark RAVE against Fourier Domain Adaptation (FDA) [15] and Progressive Cyclic Style-Transfer (ProCST) [16] to compare visual fidelity and style similarity. This experimental setup enables a precise comparison, allowing us to evaluate whether the features such as appearance and style of the target images are successfully embedded into the source data without distorting or eliminating the content of the source images. Our experiments demonstrate that RAVE, with precise text prompts, surpasses FDA and ProCST in terms of FVD scores.

In summary, our contributions are twofold. First, we conducted a comprehensive review of multifaceted generative model architectures, including VAEs, GANs, and Diffusion, examining their applicability for video data augmentation — a finding summarized in a Table for future reference and comparative analysis (see appendix). Second, we leverage a diffusion-based framework to generate domain-adapted videos, advancing the capability of synthetic-to-real domain transfer by aligning GTA5's visual style more closely with Cityscapes. This work accentuates the efficacy of diffusion models in augmenting the fidelity and diagnostic utility of clinical training data and facilitates the generation of data for computer-aided detection (CAD) algorithm training. We conclude that diffusion models exhibit exemplary performance for video augmentation.

## II. Related work

In this section, we review related work on various generative models, including VAEs, GANs, and diffusion models, with a particular focus on their editing capabilities. Additionally, we discuss their applications in data augmentation to establish a robust foundation for understanding the efficiency and effectiveness of using generative models to create synthetic data for training downstream tasks.

### *VAE-based*

Variational Autoencoders (VAEs) represent a generative model that unites probabilistic inference with neural networks to encode complex data into a continuous latent space [17]. By optimizing the variational lower bound or evidence lower bound (ELBO), VAEs approximate posterior distributions and enable efficient sampling from the latent space. ST-VAE [18] is an efficient latent space-based style transfer method capable of merging multiple distinct styles into a content image. This is achieved by incorporating a Variational Linear Transformation (VLT) module in the bottleneck, which learns the covariance of both input and style features. Notably, a Variation module projects nonlinear features into a multivariate Gaussian latent space. Interpolating data in this way facilitates more seamless style transitions while maintaining the structural integrity of the content images.

### *GAN-based*

Generative Adversarial Networks (GANs) marked a pivotal milestone in the progress of deep learning, driving forward new possibilities in areas like image generation, style transfer, and video generation. In this section, we discuss notable GAN-based approaches specifically designed for video editing and style transfer. MoCoGAN [19] is designed for generating short video clips where the same object can perform different actions, or different objects can perform the same action. The model begins by generating a random vector that is decomposed into two parts: content and motion. These parts are then tokenized for the image generators. The output is assessed by both an image discriminator and a video discriminator. Unlike earlier methods, MoCoGAN does not constrain video generation to a fixed length, allowing it to handle variations in speed within a single clip more effectively. For the text-driven video generation, IRC-GAN [20] utilizes 2D convolutional networks instead of 3D networks to produce high-definition videos and strengthens temporal coherence using ConvLSTM [21]. The combination of these components embedded within the GAN architecture ensures that IRC-GAN generates high-quality visual content that is aligned with the given text prompt. Building upon the aforementioned innovative applications of GANs in video generation, other research has explored the versatility of GANs in various contexts. Zhu *et al*. [22] proposed CycleGAN for unpaired image-to-image translation. This model relies on two autoencoders, or ”bijective” functions, to learn mappings between two domains, each with its discriminator to ensure the generated images appear indistinguishable from those in the target domain. The key innovation of CycleGAN is the introduction of a cycle consistency loss, which helps enforce translating an image to the target domain and back again to the original image. This method has proven highly effective in tasks like artistic collections style transfer. StyleGAN [23] introduced a revolutionary generator architecture that resolves the problem of entangled representations via hierarchically modifying the latent code. Since then, many StyleGAN-based approaches have emerged. For example, DRB-GAN [24] decomposes the generator into two networks: a style encoding network and a style transfer network. The style transfer network utilizes a tailored module called Dynamic ResBlock, which performs integration with "style code" containing style features produced by the style encoding network. These instance features will be then fed into the style transfer network to complete the artistic style transfer. StyleGAN's adaptability extends to video editing as well. Stitch it in Time [25] leverages a pretrained StyleGAN model to edit talking-head videos and fine-tune frames by using Pivotal Tuning Inversion.

### *Diffusion-based*

Make-A-Video [26] builds upon text-to-image (T2I) diffusion models and extends it to text-to-video generation, incorporating spatial super-resolution and interpolation models to enhance video quality. One of its benefits is that it circumvents the requirement for training on paired text-video data. Imagen [27] proposes a text-driven cascaded diffusion video model capable of generating high-definition videos with strong temporal consistency. It consists of a pretrained frozen text encoder, a base video diffusion, and spatial and temporal super-resolution models. To reduce the amount of computation, it uses progressive distillation [28] which enables fast sampling of diffusion models. Tune-A-Video [29] introduces sparse spatio-temporal attention which only calculates the first and the preceding frame instead of computing attention across all frames. However, it still requires substantial effort to fine-tune the models, and editing videos with multiple objects remains challenging.

Text2Video-Zero [10] exploits a pretrained Stable Diffusion model (SD) [30] to achieve zero-shot synthesis using post-hoc strategies, such as motion dynamics encoding, to tackle temporal consistency. It further improves temporal and structural coherence with cross-frame attention, initial frame integration, and background smoothing. Despite being effective in handling static backgrounds, Text2Video-Zero faces challenges with complex dynamic backgrounds.

Dreamix [31] is a general video editing model and trained on a large-scale video-text dataset. The models initially extract features from input low-resolution video corrupted by additive Gaussian noise, and use the text prompt as guidance to map the features with corresponding original videos. To increase fidelity, Dreamix employs a tailored fine-tuning approach for each input video. Specifically, it masks temporal attention during the training phase to enhance the reconstruction. Although Dreamix can perform sophisticated video editing and offers flexibility by accepting either images or videos as input, extensive fine-tuning, being sensitive to hyperparameter selections, and demands for substantial computational resources are problematic for this model. Pix2Video [13] utilizes a pretrained structure-guided T2I model as the backbone to edit on an anchor frame, employing Denoising Diffusion Implicit Model (DDIM) inversion and depth-conditioned maps as additional structure channel. Coherent appearance is secured by selecting a reference frame, injecting its self-attention features into the U-Net, and to further improve the temporal stability, it progressively updates the latent variable at each diffusion step, enforcing the similarity for each consecutive frames.

FateZero [32] stores both spatial-temporal self-attention and cross-attention as source maps at every step of the inversion process. Then at each denoising step, the saved source maps will be fused with target editing attention maps through blending, which could preserve motion and structural information. Moreover, it incorporates a spatial-temporal attention module inspired by the causal self-attention to better address temporal consistency. Rerender [11] focuses on maintaining temporal coherence across frames. It achieves this by adapting hierarchical cross-frame constraints. Global style consistency is ensured by a sparse causal cross-frame attention module similar to Text2Video-Zero. To compensate for the insufficiency of cross-frame attention, the author uses optical flow to perform a fusion of latent features where the previous frame serves as a reference for the next frame during early denoising steps. Due to the reliance on optical flow, Rerender encounters limitations in maintaining style over longer video sequences and only performs best in scenarios with slow movements.

TokenFlow [12] presents a framework that leverages prior knowledge of the source video to inject features through propagation to neighboring frames. The first stage involves extracting spatial coordinates and corresponding latent features from the original video. Then, keyframes are sampled and processed using a spatio-temporal self-attention mechanism, similar to the one used in Tune-A-Video, yielding a set of tokens. These tokens are denoised independently with the aid of the pre-computed spatial coordinates. TokenFlow excels in maintaining temporal consistency, particularly in videos with complex motion and dynamic scenes. However, it struggles with handling drastic object transformations or significant alterations to the video layout. RAVE [9] facilitates style coherence by collectively editing video frames in the formation of grid batch in conjunction with ControlNet [33]. This approach strengthens consistent styling across frames when performing a DDIM inversion along with condition extraction via off-the-shelf methods such as depth maps to guide ControlNet, thereby enhancing stylistic consistency. However, while the grid-based approach establishes a foundation for style uniformity, it alone cannot maintain spatio-temporal integrity across video. To address this problem, RAVE introduces randomized noise shuffling, a novel technique that randomly permutes frames within the grid at the end of each denoising step. This permutation technique enhances the spatio-temporal interactions through the model's self-attention layers, reducing flickering artifacts and fostering global coherence in longer videos.

VideoComposer [34] is a multi-modal guided video generation framework that allows a diverse range of inputs, such as text descriptions, sketches, reference videos, or hand-craft motion paths. Its high level of controllability in video synthesis results from the fusion of three distinct types of conditions: textual, spatial, and temporal. In particular, a motion vector, which encodes directional information between two neighboring frames, is deployed as a signal to guide the temporal dynamics of the video. Furthermore, a Spatio-Temporal Condition encoder (STC-encoder) leveraging cross-frame attention mechanisms is devised to enhance consistency across frames through the injection of semantic image embeddings [35].

MoonShot [36] is designed to generate long-term and temporally coherent videos and can be repurposed for various applications, including animation and video editing. Its multimodal video blocks can accommodate text and image conditions with the help of decoupled multimodal cross-attention and the use of a pretrained ControlNet module for geometry conditioning.

*Applications*

The advancements in generative models, particularly GANs and diffusion models, have enabled the creation of high-quality synthetic data to enhance and alter real-world datasets. These models have demonstrated their capacity to bridge domain gaps [7], [37]–[39] secure data privacy [40], and facilitate knowledge distillation through augmented dataset [2]–[5], [41]. In domain adaptation, such models mitigate domain shifts through style transfer, making source data visually resemble the target data in preparation for downstream tasks, such as image segmentation [16], [37], [38] and stereo matching [39]. CyCADA [37] achieves image-to-image translation through pixel, feature, semantic, and cycle-consistency losses, enabling structural and semantic transformation. Building upon this, such architecture is beneficial for preserving semantic information, Li *et al*. [38] introduce bidirectional learning for semantic segmentation, where image translation and segmentation models reciprocally feedback on another in a closed-loop training process. Addressing domain discrepancies induced by variations in MR scanners across medical centers, Gong *et al*. [7] propose a conditional diffusion model, Diffuse-UDA, which reconstructs annotated MR images by utilizing paired image-mask data from both annotated source domains and pseudo-labeled target domains. Furthermore, Diffuse-UDA not only enhances segmentation accuracy, but also generates synthetic data for robust data augmentation. Beyond domain adaptation, extensive research has focused on effective data augmentation, exploring how applicable synthetic data can improve the performance of training vision tasks. It has been established that dataset diversity is pivotal, and a combination of real and synthetic data yields the optimal results for ameliorating robustness against distribution shifts [2]–[5], [41]. The healthcare sector, in particular, stands to benefit from synthetic data due to the high cost of annotation, privacy concerns, and quality assurance challenges. For instance, conditional GAN models have been developed to generate histopathology, with their reliability assessed in classification tasks [8], [42]. Unlike conventional mindset that directly augment datasets, AugDiff [43] orchestrates augmentation at the feature level, enabling the model to learn more informative and representative features. Their comprehensive experiments illustrate that this feature-level augmentation markedly enhances performance while also reducing training time due to the computational efficiency of feature-level manipulation.

## III. METHODOLOGY

*Video Augmentation for Domain Adaptation (DA)*

This paper focuses on data augmentation to achieve domain adaptation, in which the model learns to transfer the feature from the image $I_s$ on the source domain $D^s = \{(x_i^s, y_i^s) \sim P_s(x^s, y^s)\}_{i=1}^{N_s}$ to the image $I_t$ on the target domain $D^t = \{(x_i^t, y_i^t) \sim P_t(x^t, y^t)\}_{i=1}^{N_t}$. In this early work, we focus on RGB images ($x^s \in \mathbb{R}^{H\times W\times 3}$) and future work will extend the investigation to the medical domain. The goal of this paper is to develop a method that can augment the source domain to be indistinguishable from target domain in terms of visual appearance such that $P_s \approx P_t$, but without changing the semantic content of source data. Once such distribution is derived, it can serve as a data generator to produce augmentation data for downstream tasks such as segmentation and classification.

*Fourier Domain Adaptation (FDA)*

Fast Fourier Transform (FFT) is implemented to perform style transfer, which does not require any training [15]. Suppose an image has phase $\phi$ and amplitude $A$ and the Fourier transform is denoted as $\mathcal{F}$. Fourier Domain Adaptation adapts the source image by replacing the amplitude belonging to the low-frequency component with the corresponding part from the target image. Since only the low amplitude part is switched and then inverted back to the image, the content of the image remains unchanged and it now has the Low Frequency (LF) components of the target. The core formulation is as follows:

$$I_{SiT} = \mathcal{F}^{-1}[\mathcal{F}(\beta A_t^{\text{low}} + (1-\beta)A_s), \mathcal{F}\phi_s] \tag{1}$$

where "Source in Target" (SiT) represents the new transformed image and $\beta$ is a value between 0 and 1, representing the percentage of replacement. The higher $\beta$, the more components of the target image are embedded into the source image, and meanwhile introducing more artifacts. This result is shown in Fig. 4 for $\beta = 0.01$, 0.1, and 0.25, respectively.

*Instance Retrieval*

Since FDA swaps the LF components of the target domain with those of the source domain, it can be inferred that using highly dissimilar or random images may introduce artifacts or result in augmented source images of poor quality. To address this, it is preferable to swap the frequency components of images that share similar semantic content. Accordingly, we implemented a deep learning-based image retrieval method using DINOv2 [44]. Specifically, we extract features from two datasets (GTA5 and Cityscapes) and compute similarity scores based on their dot product. Then by using a search algorithm, we identify the most and least similar instances. Through this process, each GTA5 frame is matched with a reference image from Cityscapes that is sufficiently plausible to be considered its closest style reference. Let the input image be $I$ and the feature extraction function be $f(I)$. We can obtain the feature vectors for all images: $\mathbf{v} = f(I)$, where $\mathbf{v} \in \mathbb{R}^{768}$. The similarity between GTA5 and Cityscapes features is computed using the dot product in the latent space, defined as:

$$s(\mathbf{v}_i^{\text{GTA}}, \mathbf{v}_j^{\text{City}}) = \mathbf{v}_i^{\text{GTA}} \cdot \mathbf{v}_j^{\text{City}} \tag{2}$$

where $s(\mathbf{v}_i^{\text{GTA}}, \mathbf{v}_j^{\text{City}})$ denotes the similarity score between the GTA5 feature vector $\mathbf{v}_i^{\text{GTA}}$ and the Cityscapes feature vector $\mathbf{v}_j^{\text{City}}$. In this form, higher values of $s$ indicate greater similarity between the features. The resulting similarity matrix $S_i$ is then constructed, which consists of the labels of the top $k$ closest Cityscapes images and the corresponding similarity scores, represented as:

$$S_i = \begin{bmatrix} I_{j_1}^{\text{City}} & I_{j_2}^{\text{City}} & \cdots & I_{j_k}^{\text{City}} \\ s(\mathbf{v}_i^{\text{GTA}}, \mathbf{v}_{j_1}^{\text{City}}) & s(\mathbf{v}_i^{\text{GTA}}, \mathbf{v}_{j_2}^{\text{City}}) & \cdots & s(\mathbf{v}_i^{\text{GTA}}, \mathbf{v}_{j_k}^{\text{City}}) \end{bmatrix} \tag{3}$$

*RAVE*

RAVE is selected as a representative for diffusion-based state-of-the-art (SOTA) methods. Fig. 3 visualizes its architecture. In the preprocessing stage, RAVE performs condition image preprocessing, which in our case utilizes a *lineart*-realistic approach, turning input video frames into *linearts* for ControlNet reference. In parallel, it obtains the latent representation for the given input video from a DDIM inversion along with a pretrained text-to-image diffusion model for efficiency and control. In the video editing stage, RAVE uses Latent Diffusion Models (LDMs), which have a U-Net as the backbone. Inside the U-Net, there are residual blocks and transformer blocks. Furthermore, ControlNet is integrated to process conditions, latent inputs, and the target text prompt. This control mech-

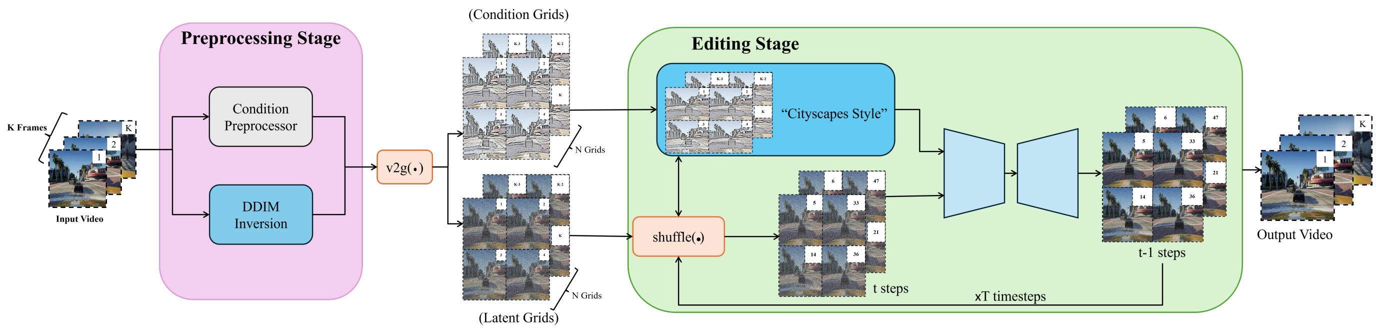


Fig. 3. Framework of RAVE. The process begins with a DDIM inversion using the pretrained T2I model and a condition extraction via a condition preprocessor. v2g function is to segment the frames into grid batch. During the editing stage, diffusion denoising is carried out for $T$ timesteps using condition grids, latent grids, and the target text prompt as inputs for ControlNet. At each denoising step, the latent grids and condition grids undergo random shuffling.

anism enhances controllability and stability throughout the denoising process.

Two methods that don’t require training are used in RAVE:

- **Grid Trick:** In the preprocessing phase, multiple frames are processed as a single image to enhance global consistency. Presumably we have grid size $\mathcal{S} = x \times y$, $N_{th}$ total number of grids and $\mathcal{P} = K - (S+1)$. The performance of this method depends on the amount of available GPU memory. The function for the grid trick is expressed as follows:
$$v2g = \left\{ G_1^{1,\ldots,S}, \ldots, G_N^{P,\ldots,K} \right\} \tag{4}$$
- **Noise Shuffling:** In the editing block, instead of processing the grid batch one after one, RAVE randomly shuffles the sequence of frames within latent grids and conditioning grids, claiming that this mitigates structural degradation. This operation is formulated as follows:
$$\text{shuffle}(G) = \left\{ G_1^{k,\ldots,l}, \ldots, G_N^{m,\ldots,n} \right\} \tag{5}$$
where $(k, \ldots, l), \ldots, (m, \ldots, n) \in \{1, \ldots, K\}$ represents random order of the grids.

*ProCST*

ProCST (Progressive Cyclic Style-Transfer) is a multiscale image-to-image translation framework designed to bridge the domain gap between synthetic (source) and real-world (target) datasets. Drawing inspiration from CycleGAN and Progressive GAN [45], ProCST progressively trains generators using a cyclic style transfer loss to transform source domain images into ”Source in Target” (SiT) images. These images retain the semantic content of the source while adopting the appearance and style of the target. The multiscale approach in ProCST involves training a generator across multiple resolutions (scales). At each scale, the generator translates low-resolution source images into low-resolution stylized images and progressively refines the details and stylistic elements as the resolution increases. This ensures that the semantic content of the source is preserved while enhancing the fidelity of the stylized SiT images.

## IV. Experiment

*Dataset*

In this study we use GTA5 and Cityscapes datasets for evaluation, designating them as the source and target data, respectively. Fourteen clips are extracted from a GTA5 driving tour video, as the existing GTA5 dataset comprises images in random order, which is unsuitable for our video consistency requirement. The Cityscapes dataset includes two sets of 14 clips, matched to the same frames as the GTA5 clips, consisting of the most and least similar images to their corresponding GTA5 frames, identified through an instance retrieval method. All videos are configured to a duration of 4 seconds, with a resolution of 512×512 pixels.

*Implementation Setup*

All methods were executed on the Snellius platform, which provides an A100 GPU. We utilized the official repositories for FDA, ProCST, and RAVE. ProCST was implemented using its pretrained model, while RAVE employed SD 1.5 from the official Huggingface repository and employs a 3×3 grid size along with 100 inference and inversion steps.

*Qualitative assessment*

This section presents a detailed comparison of various methods, using key examples and findings to illustrate the effectiveness of RAVE, FDA, and ProCST, with a focus on style transfer and structure preservation.

Fig. 4 displays selected examples from one of the clips. RAVE 1-5 corresponds to the following prompts:

1) ”Car is driving in a cityscape-style environment.”
2) ”Make this video appear like it belongs in the Cityscapes dataset.”
3) ”Apply Cityscapes style.”
4) ”Make this video like it belongs in the Cityscapes dataset,” with negative prompt: ”do not change the color of trees.”
5) ”Car is driving in a German city,” with negative prompts: ”rain, sunny, dark night.”

RAVE utilizes Stable Diffusion (SD), and we presume that the Cityscapes dataset is part of its training data, enabling RAVE to interpret and convey the essence of Cityscapes through the prompts. Each variation in the prompts results in

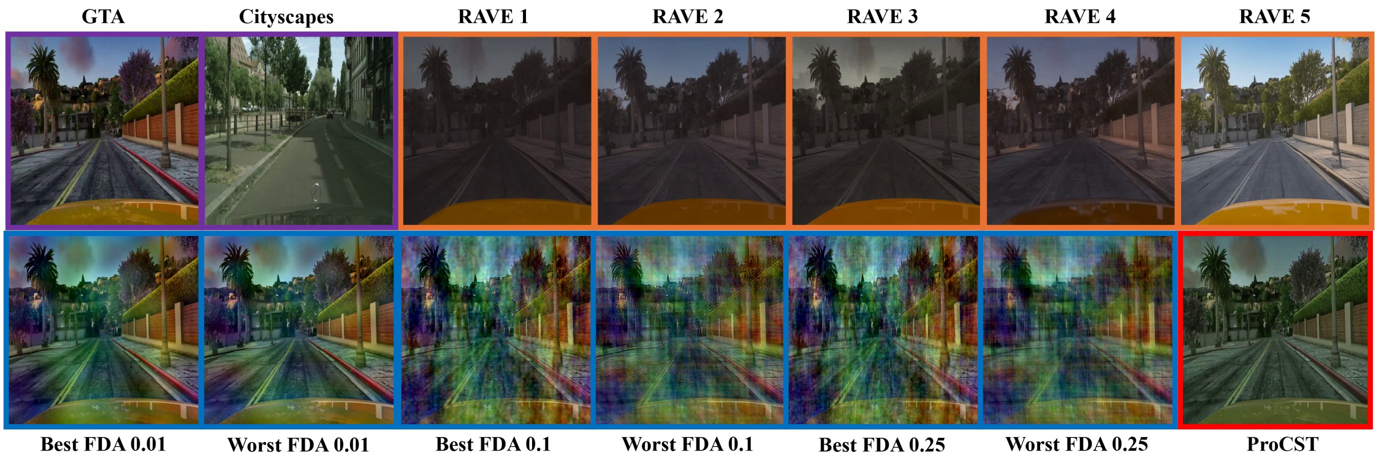


Fig. 4. Images from Video-14, its most similar Cityscapes image (obtained with instance retrieval), and examples of different style-transferred video frames. Note that in this example, the FVD scores for best and worst are very close, making the differences too subtle to be perceptible to the naked eye.


Fig. 5. The central logo on the building in the GTA5 image loses detail in RAVE-processed frames, while ProCST preserves the logo, altering only the style.

subtle stylistic differences. The outputs generally align with our expectations, presenting a gloomy and muted ambiance in RAVE 1-4. In contrast, RAVE 5, where the prompt specifies a German city and includes negative conditions to avoid bright or rainy scenes, aims to capture the overall Cityscapes style. Its visual output closely resembles the Cityscapes imagery shown in Figure 1. However, RAVE videos demonstrate a loss of detail, which is quantitatively reflected in the SSIM scores, as shown in Fig. 5.

For the FDA set, we employ instance retrieval to identify the most and least similar Cityscapes images to the given GTA image, applying FDA with three different beta parameters. A higher beta value indicates a greater transformation of Cityscapes components into the GTA5, theoretically aligning GTA images more closely with the Cityscapes style. Nevertheless, this comes at the cost of significantly introducing artifacts, potentially rendering the videos less visually clear. The distinction between the optimal and suboptimal conversions, observed at beta values of 0.1 and 0.25, demonstrates that the best conversions maintain more fineness.

For ProCST, we utilize their official pretrained model, which is tailored to capturethe nuances of Cityscapes style, rendering the images visually similar to the Cityscapes dataset while avoiding artifact introduction and preserving the fine-grained intricacies of the original GTA image, as supported empirically by quantitative results (SSIM).

*Quantitative evaluation*

Following established methodologies [26], [34], [46], we employ Fréchet Video Distance (FVD) to assess the temporal coherence, frame quality, and feature distribution of Source in Target GTA5 (SiT-GTA5) videos by comparing them to Cityscapes videos. Lower FVD scores indicate that the distribution of the generated video's features is closer to that of real video data, signifying greater realism. We also calculate the Structural Similarity Index (SSIM) [47] between the original GTA5 and SiT-GTA5 images, with higher SSIM values signifying less structural variability (higher consistency with the source image). Results are presented in Table 1 and 2. Across the 14 videos, 71.4% (10/14) of SiT-GTA5 videos in both the RAVE group and the ProCST group, and 42.8% (6/14) of SiT-GTA5 videos in the FDA group demonstrate an improvement, indicating that these three methods are effective for domain adaptation. Notably, 85.7% (12/14) of the RAVE group outperforms ProCST, with the performance difference being as much as approximately 6.7 times lower in Video-7 for RAVE Prompt 3 (0.321) compared to ProCST (2.166). Conversely, in instances where ProCST performs better than RAVE, the disparity is limited to only 3 times, as observed in Video-3 with RAVE Prompt 5 (4.356) and ProCST (1.404). Furthermore, for videos containing a higher number of frames (see Video-1, Video-2, Video-4, Video-5, Video-12, Video-13, Video-14), RAVE's performance exceeds that of ProCST, suggesting that RAVE is particularly advantageous for high-FPS videos. On the other hand, FDA falls short of our expectations, exhibiting a marked decline in performance and showing only a minor difference between best and worst similarity conversions to Cityscapes. This may be due to the introduction of noticeable artifacts by FDA, which severely impacts video quality and is reflected in higher FVD scores. Additionally, while increasing the beta parameter reduces FVD scores for some cases, a significant drop in SSIM scores is observed when beta shifts from 0.01 to 0.1. As noted in the qualitative evaluation, RAVE group's SSIM scores are on average 64 lower than ProCST's (88.64), indicating that RAVE tends to smooth out certain macroscopic textures.

Table 1. FVD scores for different methods across videos. The average FVD score is calculated from the best result per video for each method.

| Method | | FVD↓ | | | | | | | | | | | | | | Average FVD |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | Video-1 | Video-2 | Video-3 | Video-4 | Video-5 | Video-6 | Video-7 | Video-8 | Video-9 | Video-10 | Video-11 | Video-12 | Video-13 | Video-14 | |
| Original GTA | | 5.592 | 13.094 | 2.744 | 2.998 | 13.312 | 0.327 | 0.305 | 5.493 | 0.388 | 9.447 | 0.950 | 9.063 | 5.016 | 5.088 | 5.27 |
| FDA group | $\beta$=0.01 | 8.350 | 15.208 | 7.723 | 3.505 | 11.688 | 1.644 | 0.043 | **2.322** | 3.891 | 10.792 | 1.380 | 15.600 | 13.415 | 13.954 | 6.09 |
| | $\beta$=0.01(worst) | 6.144 | 34.990 | 15.755 | 8.886 | 19.407 | 1.358 | **0.007** | 7.219 | 4.001 | 13.048 | 1.687 | 14.029 | 12.175 | 13.729 | |
| | $\beta$ = 0.1 | 14.642 | 25.402 | 14.648 | 3.931 | 19.663 | 6.032 | 9.282 | 12.660 | 7.310 | 12.698 | 1.295 | 11.749 | 10.521 | 5.882 | |
| | $\beta$=0.1(worst) | 14.476 | 23.856 | 15.928 | 7.015 | 17.485 | 4.547 | 9.909 | 19.767 | 8.325 | 21.097 | **0.236** | 14.346 | 11.774 | 4.632 | |
| | $\beta$=0.25 | 12.201 | 24.500 | 16.175 | 3.960 | 16.272 | 5.919 | 9.338 | 10.729 | 8.897 | 15.277 | 1.422 | 7.265 | 10.840 | 5.330 | |
| | $\beta$=0.25(worst) | 15.832 | 22.109 | 18.763 | 5.527 | 14.286 | 5.715 | 13.093 | 24.456 | 8.195 | 24.701 | 0.620 | 10.732 | 12.489 | 4.957 | |
| RAVE group | prompt 1 | 2.501 | 15.079 | 8.520 | 2.912 | **3.854** | **0.391** | 1.088 | 2.759 | 2.545 | 8.016 | 0.963 | 6.981 | 6.631 | 1.369 | 2.73 |
| | prompt 2 | 2.497 | 10.982 | 5.313 | 2.191 | 5.982 | 0.863 | 1.863 | 4.639 | 1.004 | 5.273 | 0.367 | 8.222 | 10.371 | 1.395 | |
| | prompt 3 | 5.347 | 14.355 | 7.852 | 2.336 | 8.169 | 1.062 | 0.321 | 7.325 | 0.791 | 6.519 | 2.616 | 6.627 | **1.822** | **1.361** | |
| | prompt 4 | **1.990** | 10.020 | 9.889 | **1.056** | 7.814 | 0.488 | 1.618 | 5.205 | 1.411 | **3.352** | 0.707 | **4.737** | 4.269 | 2.105 | |
| | prompt 5 | 5.504 | **6.465** | 4.356 | 1.476 | 13.528 | 0.891 | 0.546 | 6.363 | **0.487** | 3.859 | 0.826 | 12.061 | 6.068 | 2.561 | |
| ProCST | | 2.913 | 7.845 | **1.404** | 1.406 | 18.969 | 0.581 | 2.166 | 2.998 | 1.438 | 6.497 | 0.253 | 6.644 | 3.457 | 3.733 | 4.31 |
| #Frames | | 36 | 25 | 17 | 45 | 25 | 17 | 17 | 17 | 17 | 17 | 17 | 53 | 72 | 36 | — |

Table 2. SSIM scores for different methods.

| **Method** | **Property** | **SSIM**($\times 10^2$) $\uparrow$ | **Average SSIM** |
|---|---|---|---|
| **FDA group** | $\beta$=0.01 | 83.43 | 68 |
| | $\beta$=0.01(worst) | 80.13 | |
| | $\beta$=0.1 | 66.44 | |
| | $\beta$=0.1(worst) | 63.18 | |
| | $\beta$=0.25 | 57.41 | |
| | $\beta$=0.25(worst) | 54.51 | |
| **RAVE group** | prompt 1 | 59.15 | 64 |
| | prompt 2 | 63.21 | |
| | prompt 3 | 64.83 | |
| | prompt 4 | 67.68 | |
| | prompt 5 | 64.68 | |
| **ProCST** | — | 88.64 | — |

This may be attributed to the latent diffusion model's (LDM) variational autoencoder (VAE), which projects input videos into a latent space, thereby reducing some features.

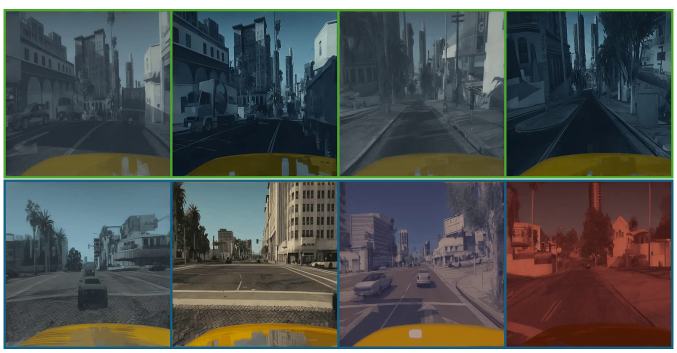

Fig. 6. Top row: Color variations within the same video. Bottom row: Different styles across various videos under the same prompt.

## V. Discussion

This section discusses several challenges encountered during the experiments and proposes potential solutions and future research directions.

Firstly, while diffusion-based generation models show aptitude for data augmentation, as indicated by favorable FVD scores and supported by existing literature, several challenges persist. One key issue is the inconsistency in color within a single video and stylistic variation across different videos, despite identical prompts (illustrated in Fig. 6). Additionally, a certain degree of fine-grained detail loss occurs (see Fig. 5). These two limitations are common across most SOTA video generative models. The underlying causes of these issues may stem from the fact that the diffusion models currently in use are not tailored for task-specific applications. This could potentially be mitigated by training conditional diffusion models designed explicitly for specialized domains, such as medical applications. To exemplify, Diffuse-UDA [7] built a conditional diffusion model from scratch to generate CT brain images, while SurgicaL-CD [6] fine-tuned the Stable Diffusion model to produce high-fidelity, diverse anatomical images. These approaches demonstrate the efficacy of diffusion models in generating consistent styles and preserving detailed textures, addressing the challenges of style inconsistency and texture loss. Another concern involves the reliability of the FVD metric. The study in [48] suggests that FVD alone may not provide a robust evaluation due to limitations in the Inflated 3D ConvNet (I3D) feature space and its dependency on a sufficiently large sample size for reliable measurement. To enhance evaluation accuracy, relying solely on FVD should be avoided. Instead, incorporating complementary metrics could yield a more comprehensive assessment. Alternatively, an indirect evaluation approach, conducted in FDA, could be adopted: using synthetic data to train a segmentation model and measuring the model's performance as an indicator of the generated data's realism, considering our primary research question centers data augmentation models for video domain adaption.

## VI. Conclusion

In this study, we investigate the potential of various models for structure-preserving video augmentation. Specifically, we select FDA, ProCST, and RAVE for experimental comparison. Through qualitative and quantitative assessments, we demonstrated that RAVE can produce high-fidelity videos, with an average FVD of 2.73, aligning closely with the target domain style. Our findings highlight diffusion models as effective tools for video data augmentation. Despite these promising results, challenges remain, particularly in maintaining consistent color and style across frames and achieving stable metrics for evaluation. Future work should focus on developing task-specific diffusion models and exploring complementary evaluation metrics to further enhance the applicability and robustness of diffusion models in various fields.

## Acknowledgment

The author is deeply grateful to Associate Professor Dr. Ir. Fons van der Sommen and Christiaan G.A. Viviers and

Francisco Caetano for their insightful feedback and guidance throughout the duration of the internship.

## References

[1] J. Song, C. Meng, and S. Ermon, “Denoising diffusion implicit models,” *arXiv preprint arXiv:2010.02502*, 2020.

[2] M. B. Sarıyıldız, K. Alahari, D. Larlus, and Y. Kalantidis, “Fake it till you make it: Learning transferable representations from synthetic imagenet clones,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 8011–8021, 2023.

[3] J. Shipard, A. Wiliem, K. N. Thanh, W. Xiang, and C. Fookes, “Diversity is definitely needed: Improving model-agnostic zero-shot classification via stable diffusion,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 769–778, 2023.

[4] B. Trabucco, K. Doherty, M. Gurinas, and R. Salakhutdinov, “Effective data augmentation with diffusion models,” *arXiv preprint arXiv:2302.07944*, 2023.

[5] R. He, S. Sun, X. Yu, C. Xue, W. Zhang, P. Torr, S. Bai, and X. Qi, “Is synthetic data from generative models ready for image recognition?,” *arXiv preprint arXiv:2210.07574*, 2022.

[6] D. K. Venkatesh, D. Rivoir, M. Pfeiffer, and S. Speidel, “Surgical-cd: Generating surgical images via unpaired image translation with latent consistency diffusion models,” *arXiv preprint arXiv:2408.09822*, 2024.

[7] H. Gong, Y. Wang, Y. Wang, J. Xiao, X. Wan, and H. Li, “Diffuse-uda: Addressing unsupervised domain adaptation in medical image segmentation with appearance and structure aligned diffusion models,” *arXiv preprint arXiv:2408.05985*, 2024.

[8] R. J. Chen, M. Y. Lu, T. Y. Chen, D. F. Williamson, and F. Mahmood, “Synthetic data in machine learning for medicine and healthcare,” *Nature Biomedical Engineering*, vol. 5, no. 6, pp. 493–497, 2021.

[9] O. Kara, B. Kurtkaya, H. Yesiltepe, J. M. Rehg, and P. Yanardag, “Rave: Randomized noise shuffling for fast and consistent video editing with diffusion models,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 6507–6516, 2024.

[10] L. Khachatryan, A. Movsisyan, V. Tadevosyan, R. Henschel, Z. Wang, S. Navasardyan, and H. Shi, “Text2video-zero: Text-to-image diffusion models are zero-shot video generators,” in *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pp. 15954–15964, 2023.

[11] S. Yang, Y. Zhou, Z. Liu, and C. C. Loy, “Rerender a video: Zero-shot text-guided video-to-video translation,” in *SIGGRAPH Asia 2023 Conference Papers*, pp. 1–11, 2023.

[12] M. Geyer, O. Bar-Tal, S. Bagon, and T. Dekel, “Tokenflow: Consistent diffusion features for consistent video editing,” *arXiv preprint arXiv:2307.10373*, 2023.

[13] D. Ceylan, C.-H. P. Huang, and N. J. Mitra, “Pix2video: Video editing using image diffusion,” in *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pp. 23206–23217, 2023.

[14] T. Unterthiner, S. Van Steenkiste, K. Kurach, R. Marinier, M. Michalski, and S. Gelly, “Towards accurate generative models of video: A new metric & challenges,” *arXiv preprint arXiv:1812.01717*, 2018.

[15] Y. Yang and S. Soatto, “Fda: Fourier domain adaptation for semantic segmentation,” in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pp. 4085–4095, 2020.

[16] S. Ettedgui, S. Abu-Hussein, and R. Giryes, “Procst: Boosting semantic segmentation using progressive cyclic style-transfer,” *arXiv preprint arXiv:2204.11891*, 2022.

[17] D. P. Kingma, “Auto-encoding variational bayes,” *arXiv preprint arXiv:1312.6114*, 2013.

[18] Z.-S. Liu, V. Kalogeiton, and M.-P. Cani, “Multiple style transfer via variational autoencoder,” in *2021 IEEE International Conference on Image Processing (ICIP)*, pp. 2413–2417, IEEE, 2021.

[19] S. Tulyakov, M.-Y. Liu, X. Yang, and J. Kautz, “Mocogan: Decomposing motion and content for video generation,” in *Proceedings of the IEEE conference on computer vision and pattern recognition*, pp. 1526–1535, 2018.

[20] K. Deng, T. Fei, X. Huang, and Y. Peng, “Irc-gan: Introspective recurrent convolutional gan for text-to-video generation.,” in *IJCAI*, pp. 2216–2222, 2019.

[21] X. Shi, Z. Chen, H. Wang, D.-Y. Yeung, W.-K. Wong, and W.-c. Woo, “Convolutional lstm network: A machine learning approach for precipitation nowcasting,” *Advances in neural information processing systems*, vol. 28, 2015.

[22] J.-Y. Zhu, T. Park, P. Isola, and A. A. Efros, “Unpaired image-to-image translation using cycle-consistent adversarial networks,” in *Proceedings of the IEEE international conference on computer vision*, pp. 2223–2232, 2017.

[23] T. Karras, S. Laine, and T. Aila, “A style-based generator architecture for generative adversarial networks,” in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pp. 4401–4410, 2019.

[24] W. Xu, C. Long, R. Wang, and G. Wang, “Drb-gan: A dynamic resblock generative adversarial network for artistic style transfer,” in *Proceedings of the IEEE/CVF international conference on computer vision*, pp. 6383–6392, 2021.

[25] R. Tzaban, R. Mokady, R. Gal, A. Bermano, and D. Cohen-Or, “Stitch it in time: Gan-based facial editing of real videos,” in *SIGGRAPH Asia 2022 Conference Papers*, pp. 1–9, 2022.

[26] U. Singer, A. Polyak, T. Hayes, X. Yin, J. An, S. Zhang, Q. Hu, H. Yang, O. Ashual, O. Gafni, *et al.*, “Make-a-video: Text-to-video generation without text-video data,” *arXiv preprint arXiv:2209.14792*, 2022.

[27] J. Ho, W. Chan, C. Saharia, J. Whang, R. Gao, A. Gritsenko, D. P. Kingma, B. Poole, M. Norouzi, D. J. Fleet, *et al.*, “Imagen video: High definition video generation with diffusion models,” *arXiv preprint arXiv:2210.02303*, 2022.

[28] T. Salimans and J. Ho, “Progressive distillation for fast sampling of diffusion models,” *arXiv preprint arXiv:2202.00512*, 2022.

[29] J. Z. Wu, Y. Ge, X. Wang, S. W. Lei, Y. Gu, Y. Shi, W. Hsu, Y. Shan, X. Qie, and M. Z. Shou, “Tune-a-video: One-shot tuning of image diffusion models for text-to-video generation,” in *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pp. 7623–7633, 2023.

[30] R. Rombach, A. Blattmann, D. Lorenz, P. Esser, and B. Ommer, “High-resolution image synthesis with latent diffusion models,” in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pp. 10684–10695, 2022.

[31] E. Molad, E. Horwitz, D. Valevski, A. R. Acha, Y. Matias, Y. Pritch, Y. Leviathan, and Y. Hoshen, “Dreamix: Video diffusion models are general video editors,” *arXiv preprint arXiv:2302.01329*, 2023.

[32] C. Qi, X. Cun, Y. Zhang, C. Lei, X. Wang, Y. Shan, and Q. Chen, “Fatezero: Fusing attentions for zero-shot text-based video editing,” in *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pp. 15932–15942, 2023.

[33] L. Zhang, A. Rao, and M. Agrawala, “Adding conditional control to text-to-image diffusion models,” in *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pp. 3836–3847, 2023.

[34] X. Wang, H. Yuan, S. Zhang, D. Chen, J. Wang, Y. Zhang, Y. Shen, D. Zhao, and J. Zhou, “Videocomposer: Compositional video synthesis with motion controllability,” *Advances in Neural Information Processing Systems*, vol. 36, 2024.

[35] F.-A. Croitoru, V. Hondru, R. T. Ionescu, and M. Shah, “Diffusion models in vision: A survey,” *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 45, no. 9, pp. 10850–10869, 2023.

[36] D. J. Zhang, D. Li, H. Le, M. Z. Shou, C. Xiong, and D. Sahoo, “Moonshot: Towards controllable video generation and editing with multimodal conditions,” *arXiv preprint arXiv:2401.01827*, 2024.

[37] J. Hoffman, E. Tzeng, T. Park, J.-Y. Zhu, P. Isola, K. Saenko, A. Efros, and T. Darrell, “Cycada: Cycle-consistent adversarial domain adaptation,” in *International conference on machine learning*, pp. 1989–1998, Pmlr, 2018.

[38] Y. Li, L. Yuan, and N. Vasconcelos, “Bidirectional learning for domain adaptation of semantic segmentation,” in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pp. 6936–6945, 2019.

[39] R. Liu, C. Yang, W. Sun, X. Wang, and H. Li, “Stereogan: Bridging synthetic-to-real domain gap by joint optimization of domain translation and stereo matching,” in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pp. 12757–12766, 2020.

[40] S.-C. S. Cheung, H. Wildfeuer, M. Nikkhah, X. Zhu, and W. Tan, “Learning sensitive images using generative models,” in *2018 25th IEEE international conference on image processing (ICIP)*, pp. 4128–4132, IEEE, 2018.

[41] H. Bansal and A. Grover, “Leaving reality to imagination: Robust classification via generated datasets,” *arXiv preprint arXiv:2302.02503*, 2023.

[42] Y. Xue, J. Ye, Q. Zhou, L. R. Long, S. Antani, Z. Xue, C. Cornwell, R. Zaino, K. C. Cheng, and X. Huang, “Selective synthetic augmentation with histogan for improved histopathology image classification,” *Medical image analysis*, vol. 67, p. 101816, 2021.

[43] L. Dai, Y. Wang, H. Wang, Y. Zhang, *et al.*, “Augdiff: Diffusion based feature augmentation for multiple instance learning in whole slide image,” *IEEE Transactions on Artificial Intelligence*, 2024.

[44] M. Oquab, T. Darcet, T. Moutakanni, H. V. Vo, M. Szafraniec, V. Khalidov, P. Fernandez, D. Haziza, F. Massa, A. El-Nouby, R. Howes, P.-Y.

Huang, H. Xu, V. Sharma, S.-W. Li, W. Galuba, M. Rabbat, M. Assran, N. Ballas, G. Synnaeve, I. Misra, H. Jegou, J. Mairal, P. Labatut, A. Joulin, and P. Bojanowski, "Dinov2: Learning robust visual features without supervision," 2023.

[45] T. Karras, T. Aila, S. Laine, and J. Lehtinen, "Progressive growing of gans for improved quality, stability, and variation. arxiv 2017," *arXiv preprint arXiv:1710.10196*, pp. 1–26, 2018.

[46] Y. Tian, J. Ren, M. Chai, K. Olszewski, X. Peng, D. N. Metaxas, and S. Tulyakov, "A good image generator is what you need for high-resolution video synthesis," *arXiv preprint arXiv:2104.15069*, 2021.

[47] Z. Wang, A. C. Bovik, H. R. Sheikh, and E. P. Simoncelli, "Image quality assessment: from error visibility to structural similarity," *IEEE transactions on image processing*, vol. 13, no. 4, pp. 600–612, 2004.

[48] G. Ya, G. Favero, Z. H. Luo, A. Jolicoeur-Martineau, C. Pal, *et al.*, "Beyond fvd: Enhanced evaluation metrics for video generation quality," *arXiv preprint arXiv:2410.05203*, 2024.